



Data Warehouse Solution for Online Furniture Retailer (Online Möbelhandel)

Course: Data Engineer

Project Participant: Pubali Mazumder

Business Context & Constraints

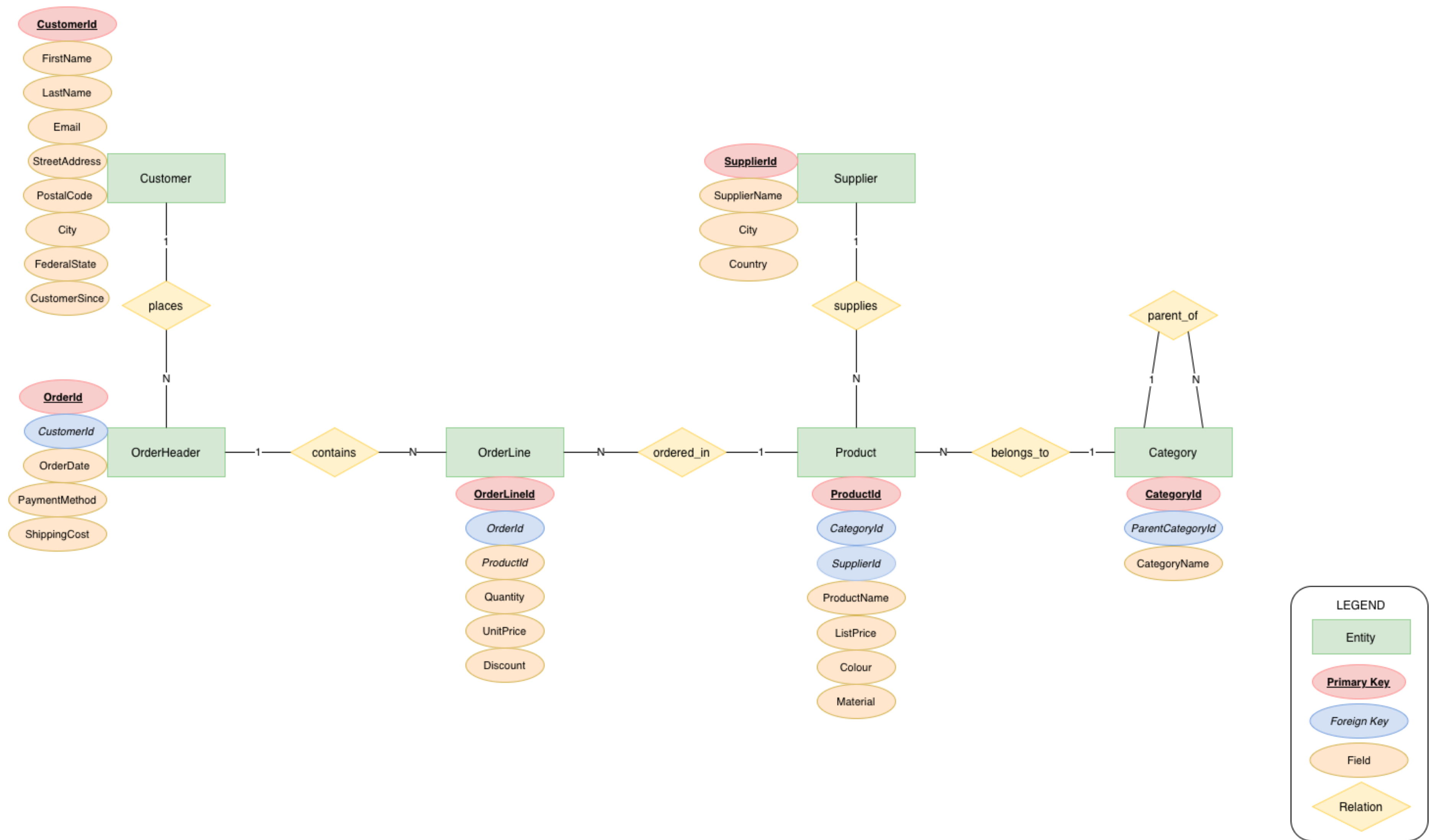
Business DB & Its Constraints

- Customer places an order (OrderHeader).
- OrderHeader has ≥ 1 OrderLine.
- OrderLine references 1 Product of a Category.
- Category belongs to only one TopCategory (Sofa to Wohnzimmer)
- Each Product has only one Supplier.

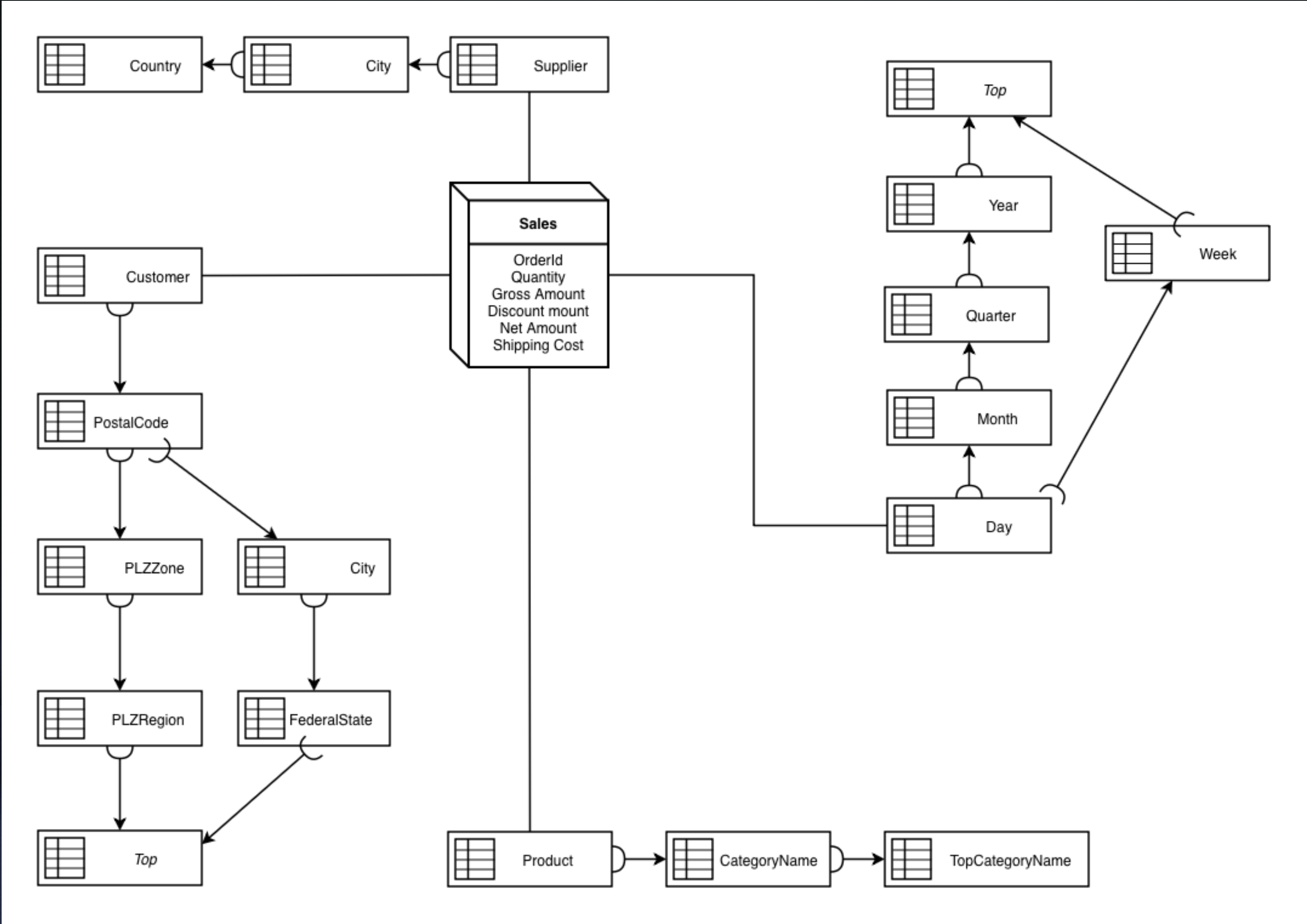
The Problem

- Operational DB optimized for single transactions, not analytics.
- Analytical queries block operational shop traffic.
- **Solution:** Separate analytical workloads into a dedicated Data Warehouse.

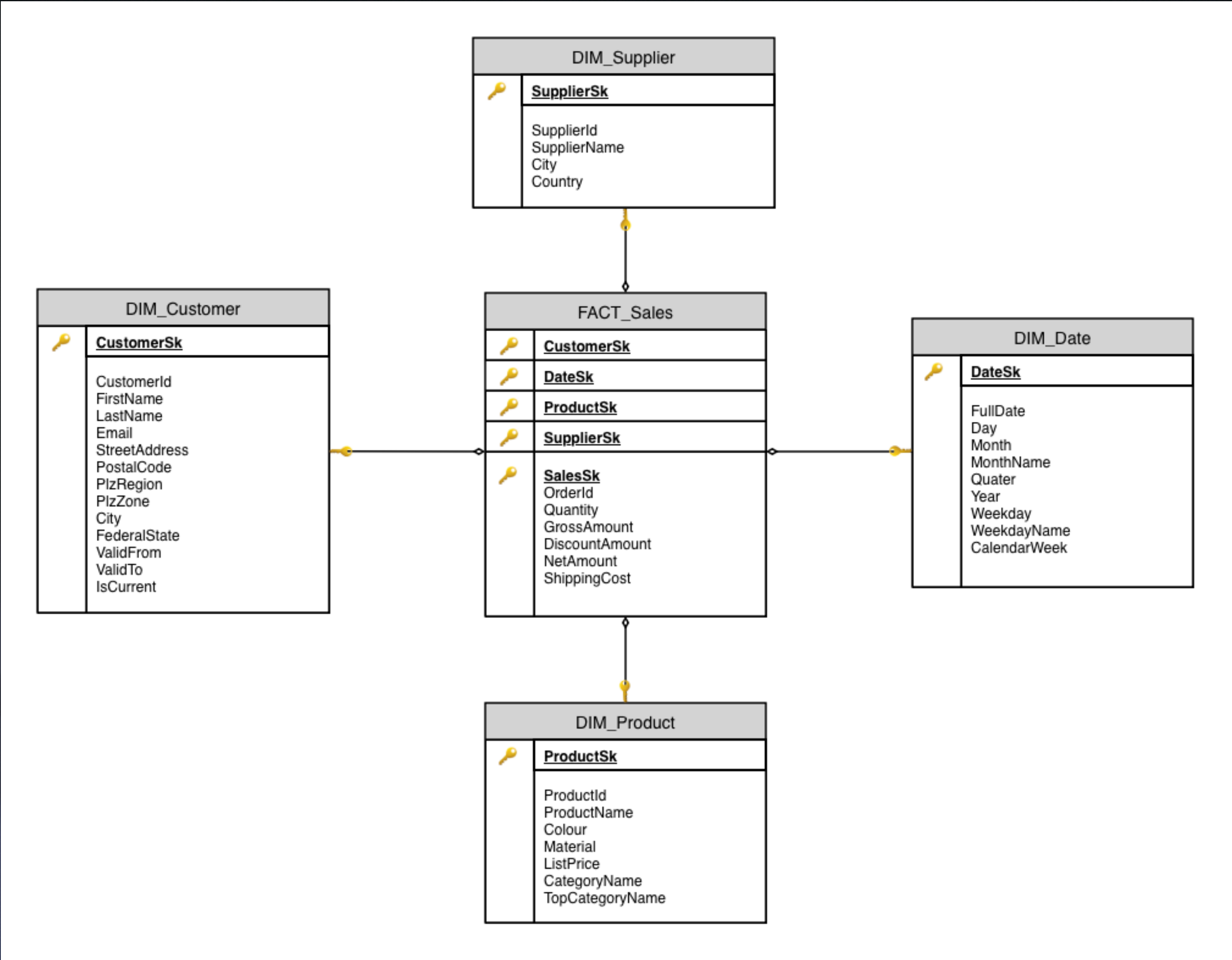
Business DB Design (ER)



DWH Conceptual Design (mER)



DWH Logical Design (Star Schema)



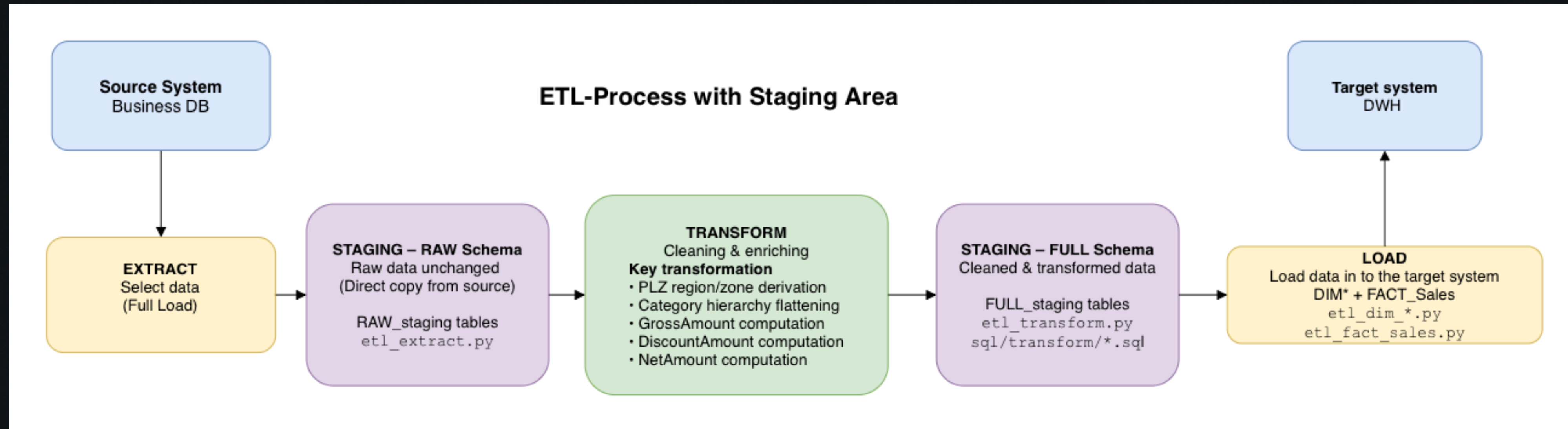
Slowly Changing Dimensions (SCD)

Dimension	SCD Type	Attributes	Justification
DIM_Date/_Supplier	Type 0	—	Immutable attributes
DIM_Product	Type 1	Name, Price, Colour	Overwrite allowed; Historical product states not required
DIM_Customer	Type 2	PostalCode, City,Street,Address, PLZRegion/Zone	Customer relocation➡ History important for analytics. Expire old row & insert new row

SCD 2 Implementation (DIM_Customer)

Tracks validity using: ValidFrom (DATE), ValidTo (DATE), and IsCurrent (0/1 flag).

Implementation & ETL Pipeline



Data Generation

- No real data set found
- Data generated using python (Faker module)

Technology Stack

- **Language:** Python 3.10+, SQL
- **Database:** SQLite3 (Serverless, Portable).
- **Schedule:** Overnight (Batch)/On-Demand

Challenges Faced & Lessons Learned

Challenge & Solution

Issue: PLZ/City mappings using Faker

Solution: Created a script to generate correct mappings

Issue: SCD update and testing

Solution: Created a script to incrementally change the DB

Lessons Learned

- Synthetic testing data needs to be reviewed for realistic analysis
- Understood the complexity of ETL even with this simple example



Thank You for Your Attention!